

Module-2

2 Marks Questions (Study Materials):

1. What is supervised learning in machine learning?
2. Name two real-world applications of supervised learning.
3. What are features and labels in supervised learning?
4. Give two advantages of supervised learning.
5. What is classification in machine learning?
6. Name any two classification algorithms.
7. Give one example of binary classification and one of multi-class classification.
8. How is classification different from regression?
9. What is the target variable type in a classification problem?
10. What is the purpose of evaluating a classification model?
11. Name various metrics used to evaluate classification models.
12. What is a confusion matrix?
13. Define precision in classification evaluation.
14. What does recall measure in classification models?
15. How does the choice of **k** affect the k-NN model?
16. Name one advantage and one disadvantage of k-NN.
17. Define Euclidean distance with proper expression.
18. Which distance metrics are commonly used in k-NN?
19. What is a decision tree in machine learning?
20. Name two criteria used for splitting nodes in decision trees.
21. Mention one advantage and one disadvantage of decision trees.
22. What is overfitting in decision trees?
23. What is the main idea behind the random forest algorithm?
24. Name two advantages of random forest over a single decision tree.
25. What does “bagging” mean in random forest?
26. How does random forest reduce overfitting?
27. What is the basic objective of SVM?
28. What is the difference between a hard margin and a soft margin in SVM?
29. What does the hyperplane represent in SVM?
30. Why k-NN is a lazy learner algorithm?

5 Marks Questions (Book Bank):

SHORT ANSWER-TYPE QUESTIONS (5 MARKS EACH)

1. What is supervised learning? Why it is called so? (5)
2. Give an example of supervised learning in a hospital industry. (5)
3. Give any three examples of supervised learning. (5)
4. What is classification and regression in a supervised learning? (5)
5. Give some examples of common classification algorithms. (5)
6. Explain, in brief, the SVM model. (5)
7. What is cost of misclassification in SVM? (5)
8. Define Support Vectors in the SVM model. (5)
9. Define kernel in the SVM model. (5)
10. What are the factors determining the effectiveness of SVM? (5)
11. What are the advantages of the SVM model? (5)
12. What are the disadvantages of the SVM model? (5)
13. Write notes on
 - (a) validation error in the k NN algorithm (5)
 - (b) choosing k value in the k NN algorithm (5)
 - (c) inductive bias in a decision tree (5)
14. What are the advantages of the k NN algorithm? (5)
15. What are the disadvantages of the k NN algorithm? (5)
16. Explain, in brief, the decision tree algorithm. (5)
17. What is node and leaf in decision tree? (5)
18. What is entropy of a decision tree? (5)
19. Define information gain in a decision tree. (5)
20. Write any three strengths of the decision tree method. (5)
21. Write any three weaknesses of the decision tree method. (5)
22. Explain, in brief, the random forest model? (5)

5 Marks Questions (Study Materials):

1. Compare bagging and boosting in supervised learning.
2. Explain the role of training data and test data in supervised learning.
3. Describe the bias-variance trade-off in supervised learning.
4. Explain overfitting and underfitting in supervised learning with suitable diagrams.
5. Compare between binary classification and multi-class classification with suitable example.
6. Explain the importance of various evaluation metrics in supervised learning models.
7. Explain various distance calculation procedure in k -NN algorithm.

8. From the following table calculate various evaluation metrics to predict COVID-19

	Predicted: Positive	Predicted: Negative
Actual: Positive	TP = 50	FN = 10
Actual: Negative	FP = 5	TN = 35

9. A factory produces paper tissues and wants to classify a new tissue sample as either "Good" or "Bad" based on two features i.e. x_1 (paper quality) and x_2 (strength). The training dataset is as follows:

X_1	X_2	Class
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

A new tissue sample has features $X_1 = 3$ and $X_2 = 7$. The goal is to classify this new sample.

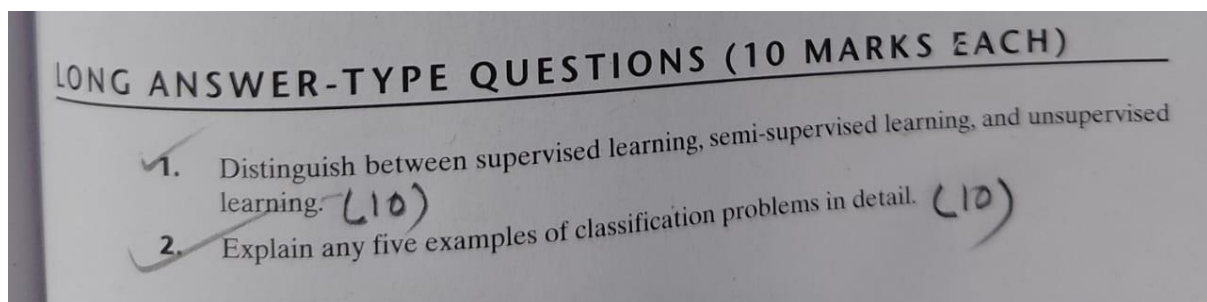
10. There is a simple 2D dataset:

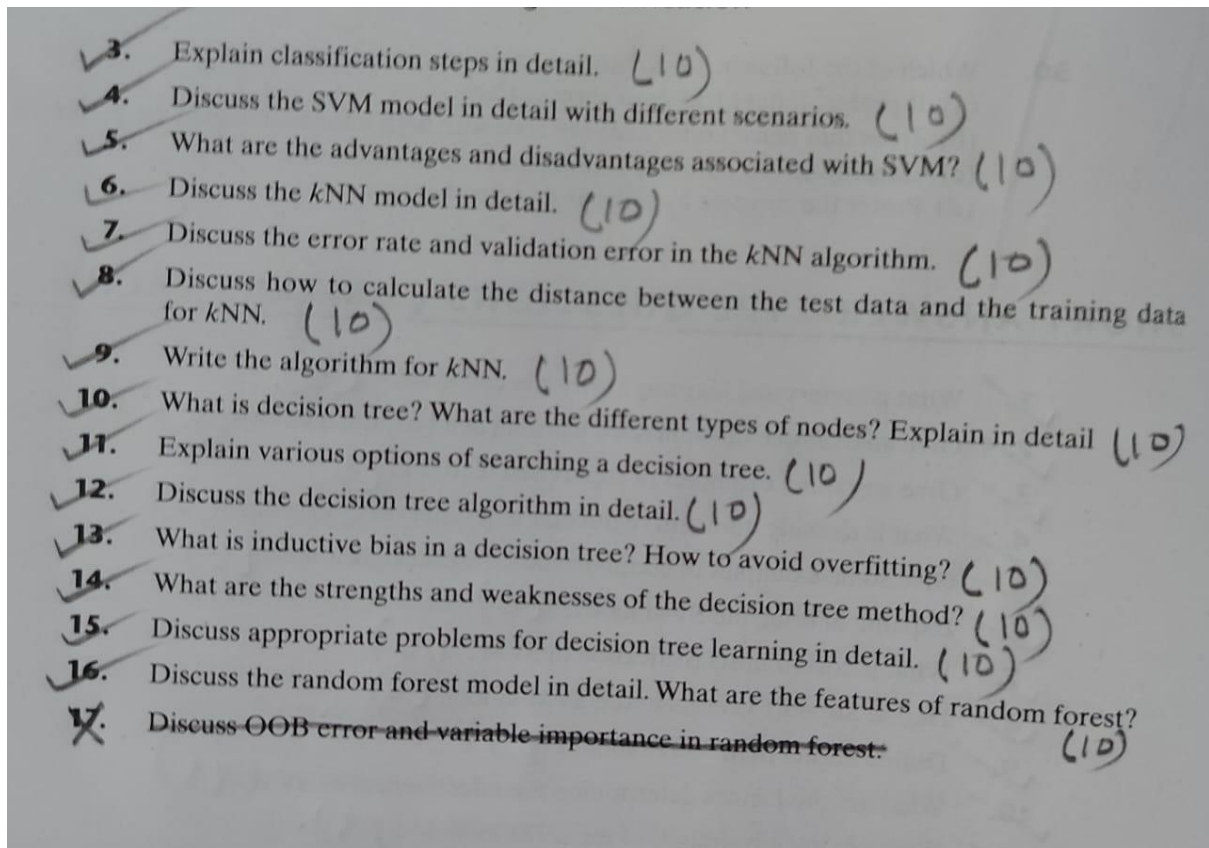
Point	X_1	X_2	Class
A	1	1	Red
B	2	3	Red
C	4	2	Blue
D	3	5	Blue

From the above table predict the class of new data point $P = (2, 2)$ using Manhattan distance, where the value of K is 3.

11. Illustrate ROC curve.
12. Compare Entropy and Information Gain in decisions tree.

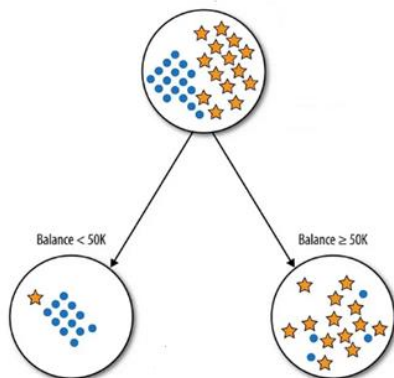
10 Marks Questions (Book Bank):





10 Marks Questions (Study Materials):

1. A company wants to predict whether a customer will buy a product based on features like age, income, and previous purchase history. Identify whether this is a classification or regression problem and describe the steps you would take to build the supervised learning model.
2. From the following figure calculate the Entropy of each node.



3. Explain K-fold cross validation with suitable example.
4. How supervised learning is used to identify a spam email, explain with suitable example.
5. Describe various challenges in supervised learning and how they can be addressed.